

29/07/2025

CUBICLE 7:

NAMES

Vidunasri R [23ecb57]

Syed Ali Fathima A [23ecb50]

Sarmatha S [23ecb28]

Vaishnavi R [23ecb54]

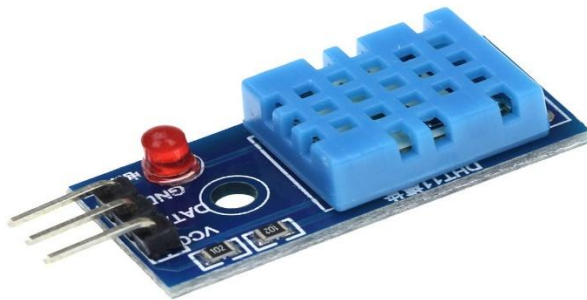
Vaishnavi S [23ecb55]

Sruthi S [23ecb45]

Trishali SJ [23ecb53]

Subagarniya M [23ecb46]

DHT11:

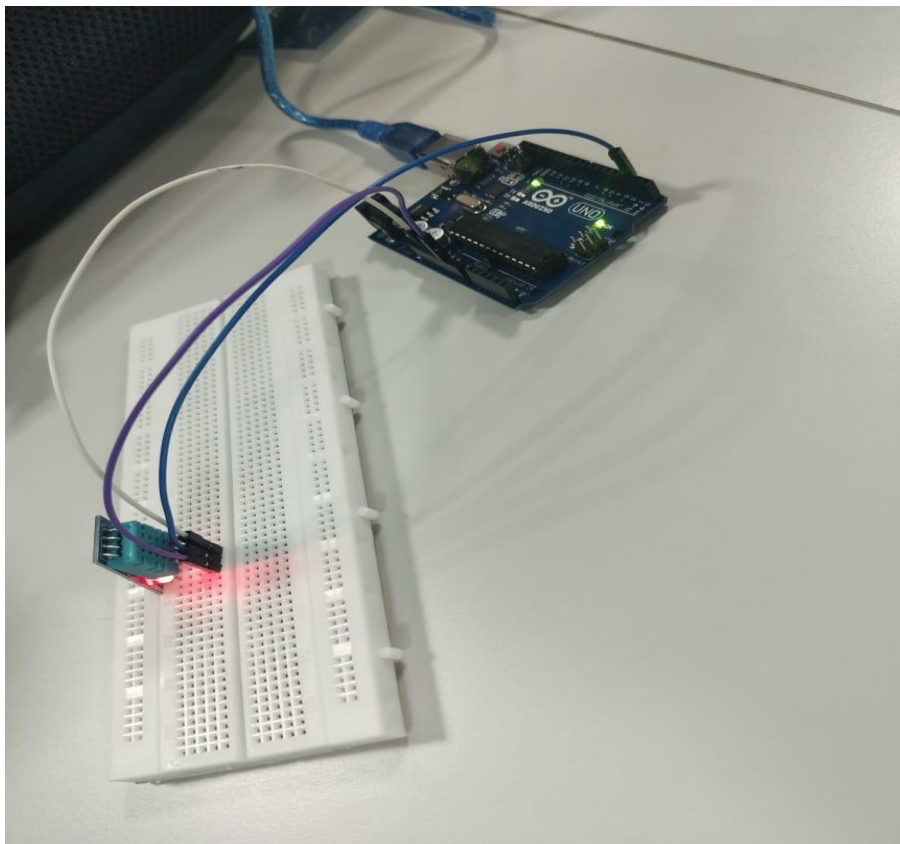


The DHT11 is a digital temperature and humidity sensor that provides reliable measurements with ease of use. It uses a capacitive humidity sensor and a thermistor to measure the surrounding air and provides the data in digital format, making it easy to interface with microcontrollers like Arduino.

It is widely used in weather monitoring, environmental sensing, greenhouse automation, and IoT projects due to its low cost, simple setup, and reasonable accuracy.

PIN DESCRIPTION:

Pin	Name	Connect To
1	VCC	3.3V or 5V
2	DATA	Digital pin (like D2)
3	GND	Ground (GND)

BOARD SETUP:**PROGRAM:**

```
#include <DHT.h>

// Define DHT11 sensor pin and type
#define DHTPIN 2
#define DHTTYPE DHT11
```

```
DHT dht(DHTPIN, DHTTYPE);

void setup() {
  Serial.begin(9600);    // Start the serial monitor
  dht.begin();           // Initialize the DHT sensor
  Serial.println("DHT11 Temperature and Humidity Monitor");
}

void loop() {
  // Wait a few seconds between measurements
  delay(2000);

  // Read humidity and temperature
  float humidity = dht.readHumidity();
  float temperature = dht.readTemperature(); // Celsius

  // Check if readings are valid
  if (isnan(humidity) || isnan(temperature)) {
    Serial.println("Error reading from DHT11 sensor!");
    return;
  }

  // Display readings
  Serial.print("Temperature: ");
  Serial.print(temperature);
  Serial.print(" °C, Humidity: ");
  Serial.print(humidity);
  Serial.println(" %");

  // Alert conditions
  if (temperature > 35) {
    Serial.println("Alert: High temperature!");
  }
}
```

```

} else if (temperature < 15) {
    Serial.println("Alert: Low temperature!");
}

if (humidity < 30) {
    Serial.println("Alert: Low humidity!");
} else if (humidity > 70) {
    Serial.println("Alert: High humidity!");
}

Serial.println("-----");
}

```

PROGRAM DESCRIPTION:

- Reads data every 2 seconds.
- Shows temperature and humidity.
- Prints alert messages if temperature or humidity is beyond safe limits.

OUTPUT:

CASE 1: ROOM TEMPERATURE:

```

sketch_Jul29a.ino
1 #include <DHT.h>
2
3 // Define DHT11 sensor pin and type
4 #define DHTPIN 2
5 #define DHTTYPE DHT11
6
7 DHT dht(DHTPIN, DHTTYPE);
8
9 void setup() {
10     Serial.begin(9600); // Start the serial monitor
11     dht.begin(); // Initialize the DHT sensor
12     Serial.println("DHT11 Temperature and Humidity Monitor");
13 }
14
15 void loop() {
16     // Wait a few seconds between measurements
17     delay(2000);
18
19     // Read humidity and temperature
20     float humidity = dht.readHumidity();
21     float temperature = dht.readTemperature(); // Celsius
22
23     // Check if readings are valid
24     if (isnan(humidity) || isnan(temperature)) {
25         Serial.println("Failed to read from DHT11 sensor!");
26     }
27
28     Serial.print("Temperature: ");
29     Serial.print(temperature);
30     Serial.print(" °C, Humidity: ");
31     Serial.print(humidity);
32     Serial.println(" %");
33 }

```

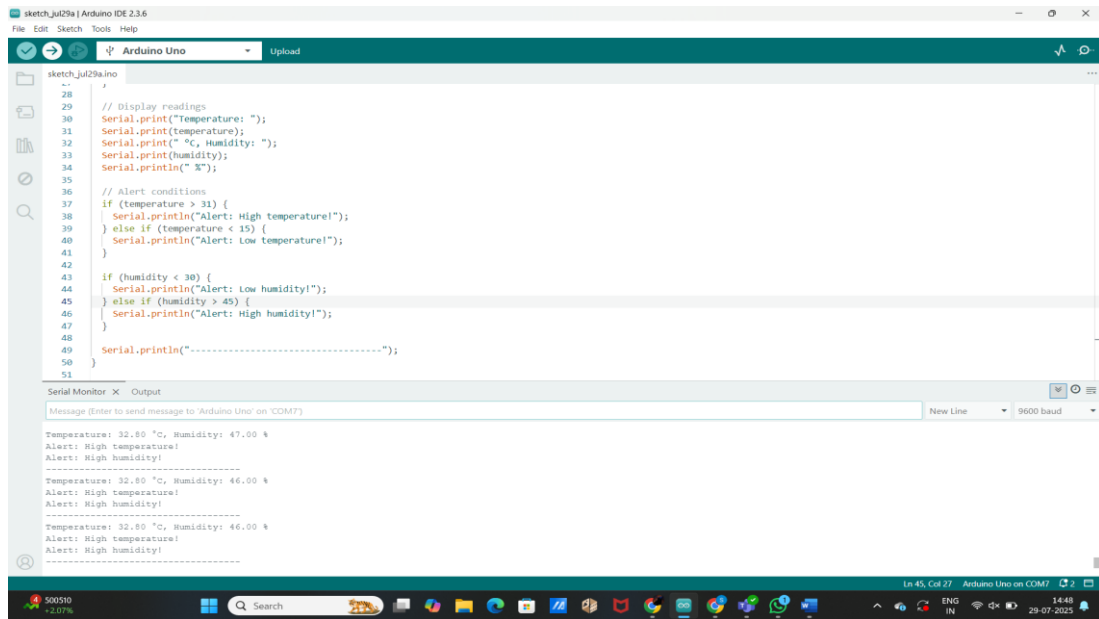
Serial Monitor X Output

```

-----
DHT11 Temperature and Humidity Monitor
Temperature: 26.20 °C, Humidity: 58.00 %
Temperature: 26.20 °C, Humidity: 58.00 %
-----
Temperature: 26.30 °C, Humidity: 58.00 %
-----
Temperature: 26.70 °C, Humidity: 58.00 %
-----
Temperature: 26.70 °C, Humidity: 58.00 %
-----

```

CASE2: HIGH TEMPERATURE:



The screenshot shows the Arduino IDE interface with a sketch named 'sketch_jul29a.ino'. The sketch code is as follows:

```
28 //
29 // Display readings
30 Serial.print("Temperature: ");
31 Serial.print(temperature);
32 Serial.print(" °C, Humidity: ");
33 Serial.print(humidity);
34 Serial.println(" %");
35
36 // Alert conditions
37 if (temperature > 31) {
38   Serial.println("Alert: High temperature!");
39 } else if (temperature < 15) {
40   Serial.println("Alert: Low temperature!");
41 }
42
43 if (humidity < 30) {
44   Serial.println("Alert: Low humidity!");
45 } else if (humidity > 45) {
46   Serial.println("Alert: High humidity!");
47 }
48
49 Serial.println("-----");
50 }
51 }
```

The Serial Monitor shows the following output:

```
Temperature: 32.80 °C, Humidity: 47.00 %
Alert: High temperature!
Alert: High humidity!
-----
Temperature: 32.80 °C, Humidity: 46.00 %
Alert: High temperature!
Alert: High humidity!
-----
Temperature: 32.80 °C, Humidity: 46.00 %
Alert: High temperature!
Alert: High humidity!
-----
```

The Windows taskbar at the bottom shows the system clock as 14:48 on 29-07-2023.